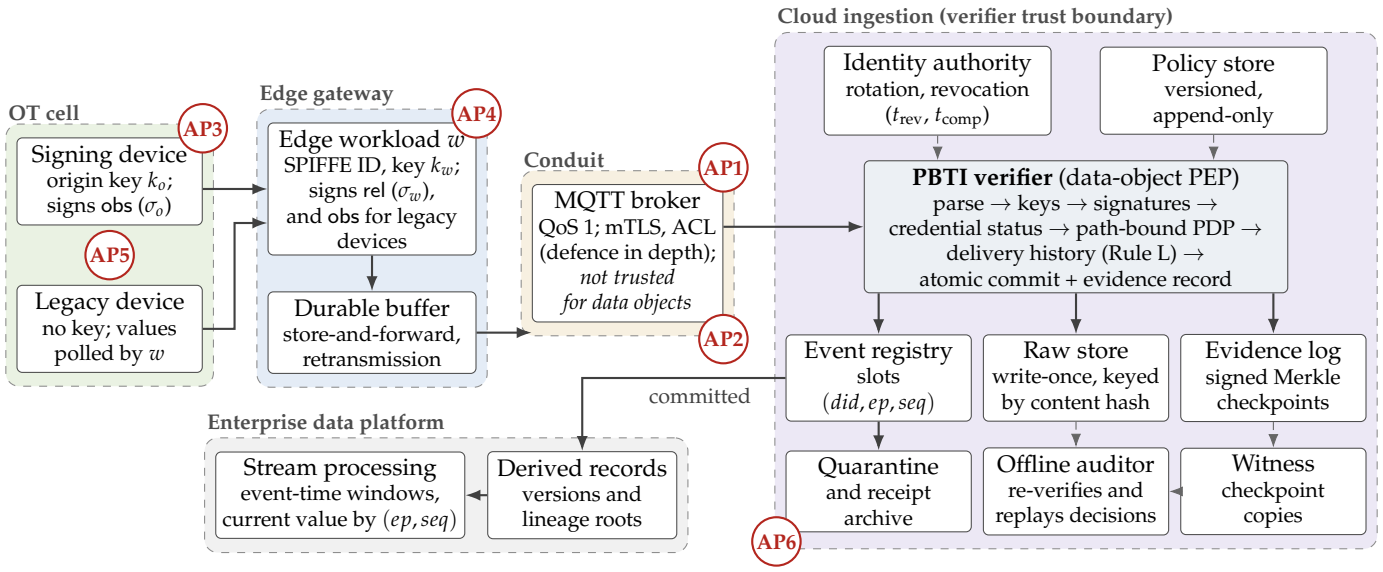

Cloud ingestion (verifier trust boundary)
OT cell
AP3
Signing device
origin key k_o;
signs obs (σ_o)
AP5
Legacy device
no key; values
polled by w
Edge gateway
AP4
Edge workload w
SPIFFE ID, key k_w;
signs rel (σ_w),
and obs for legacy
devices
Durable buffer
store-and-forward,
retransmission
Conduit
AP1
MQTT broker
QoS 1; mTLS, ACL
(defence in depth);
not trusted
for data objects
AP2
Identity authority
rotation, revocation
(t_{rev}, t_{comp})
Policy store
versioned,
append-only
PBTI verifier (data-object PEP)
parse → keys → signatures →
credential status → path-bound PDP →
delivery history (Rule L) →
atomic commit + evidence record
Event registry
slots
(did, ep, seq)
Raw store
write-once, keyed
by content hash
Evidence log
signed Merkle
checkpoints
committed
Enterprise data platform
Stream processing
event-time windows,
current value by (ep, seq)
Derived records
versions and
lineage roots
Quarantine
and receipt
archive
Offline auditor
re-verifies and
replays decisions
Witness
checkpoint
copies
AP6